

Previous game success as a determinant of future game performance and outcomes in men's NCAA basketball

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This research examines the motivational effects of past team success (or failure) as a determinant of future game performance. Analyses draw on a data base of 4,495 NCAA Division I Men's Basketball games and detailed player performances over multiple seasons. Findings show that winning percentages are strongly influenced by team success or failure in the previous game-success begets failure and failure begets success, and the effects are exacerbated by the extent of the win or loss. Contributing to the differential winning percentages among teams that either won or lost their previous games are several key basketball performance variables (shooting percentages, rebounding, turnovers, etc.) as they are reflected in both offensive and defensive efficiency ratings.

KEY WORDS: Basketball, Confidence, Efficiency Ratings, Failure, Motivation, NCAA, Outcomes success, Performance.

Motivation, Confidence and Performance

Player motivation is widely acknowledged as a critical determinant of performance and success in virtually all sports, but in basketball, a game that is so often decided by just a tenth of a second or a fraction of an inch, it is frequently concluded that player motivation – “the team that wants it more” – is what ultimately differentiates teams on the scoreboard.

One motivational factor that is recognized intuitively by coaches, players and fans is how teams performed in their most recent competition. Much is made of the perils in facing the team that was just beaten, or beaten badly. Likewise a recent win, especially an easy victory, is thought to promote overconfidence in a team and its players and to put them at higher risk for a loss in their subsequent game. Despite this ubiquitous conventional wisdom,

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empirical research has been remarkably quiet on the effects of a previous win or loss, and rarely has it been empirically examined in the sport of basketball, one of the most avidly followed and highest revenue generating team sports in North America (Schwartz 2010; Mondello, 2006) and the third most popular sport around the globe (TopEndSports, 2014).

To appreciate the importance of previous game effects on current game outcomes we consider it in the context of a larger conceptual framework, one in which game success is closely shaped by the major on-court performance variables such as shooting percentages, rebounds and turnovers. Their effects are often aggregated into two overarching composite indicators: offensive and defensive ratings (Oliver, 2004; Kubatko *et al.*, 2007). Previous research confirms that defensive rebounds (Gómez *et al.*, 2008a,b; Ibáñez *et al.*, 2008; García *et al.*, 2013), number of assists (Gómez *et al.*, 2006; Ibáñez *et al.*, 2008; García *et al.*, 2013), free throws (Sampaio and Janeira, 2003; Ibáñez *et al.*, 2003; 2008; Simovi *et al.*, 2011), and two and three-point field goals attempted (Kozar *et al.*, 1994; García *et al.*, 2013; Simovi *et al.*, 2011) are the on-court performance indicators that best differentiate between winning and losing teams.

In turn, on court performance is conditioned by two sets of off-court factors. The first set we refer to as “coach-controlled” variables such as depth of rotation, shooting/scoring balance, player size and experience (age/year in school). These are variables that, as the label suggests, are the principal levers controlled by coaches from the sidelines in their management of the game. We also note that because coach-controlled and on-court performance variables tend to change throughout the course of a game, it is important to recognize that for this reason they can be highly interactive; for example, the number of personal fouls called on a team is positively correlated with the number of players in the rotation for that game (Clay and Clay, 2014).

The second set of off-court factors is a combination of exogenous influences that we here refer to as “ecological variables.” These variables include geospatial effects such as crossing time zones (Smith *et al.*, 1997; Steenland and Deddens, 1997; Reilly and Waterhouse, 2009; Winter *et al.*, 2009; Samuels, 2012; Clay, *et al.*, 2015), playing at higher elevations (Pollard, 2006; Wilber, 2011; Nassisi, 2013) or at colder temperatures (Clay *et al.*, 2014). By far the most commonly observed exogenous influence on performance and outcomes is the benefit of playing in front of a friendly fan base, otherwise known as “home court advantage” (Greer, 1983; Smith *et al.*, 2000; Carron, 2005; Borghesi, 2007; Legaz-Arrese, 2013).

Previous game result is similarly classified as an ecological variable as it is hypothesized to affect, in a one-way causal direction, both on-court per-

formance and, in turn, current game outcomes. Several studies give insight into how motivation from a previous loss or other setback can help lead teams to victory. Most notable are the findings of Mizruchi (1991) in his study of 252 National Basketball Association (NBA) playoff games from 1947-1982. Mizruchi found that, consistent with the guiding hypotheses of the current research, the probability of success was higher for teams that had lost their previous games, and that the probability of failure was also higher for teams that had won their previous games. The focus of Mizruchi's study was on a small, specialized sample of NBA playoff games in which teams played one another repetitively in a series. He selected particular games experimentally (e.g., pairs of games from the same site) to help minimize the confounding effects of exogenous variables. To control for the reality that stronger teams tend to win their subsequent games, whether they lost the previous game or not, he developed a measure of relative team strength based on season won-loss records. Home court advantage was also controlled.

Berger and Pope (2011) analyzed over 18,000 NBA basketball games and over 45,000 NCAA basketball games and found that being behind by a small margin at halftime more often resulted in winning than in losing. While they did not examine previous game results, their research suggests that motivation and overconfidence (derived from the half-time score) may affect current game outcomes. These findings are also consistent with the "vengeful loser" and "overconfident winner" hypotheses introduced later in this section.

In a similar vein, it is well established that in the first round of the NCAA Men's Basketball Tournament the 9th seed upsets the higher 8th seed over 53% of the time (Bryan *et al.*, 2006; Baumann, *et al.*, 2010). One plausible interpretation of these tournament upsets is that the higher seed, or favored team, becomes modestly overconfident and/or the lower seed is more highly motivated to overcome the challenge of playing against a team perceived to be "superior," as reflected in its higher seed. It is possible that the same effects come into play in all of the tournament's pairings, but only in the case of the 8-9 matchup are the teams close enough in strength for the underdog to actually win a majority of the time. This effect resembles the Berger and Pope finding that only teams with a *small* half-time deficit go on to win the game. This is because teams that are close at the half tend to be evenly matched teams (like the 8-9 seeds in the NCAA Tournament). We posit that in any individual game there may be many different factors at play, resulting in a potential win or loss by the team that is down at half time. However, across thousands of games randomly distributed over time and space, the

fact that the team modestly behind at half time more often than not comes back to win suggests that there is an unmeasured influence that lies behind it. One reasonable explanation may be differences in how the teams that are up versus the teams that are down differ in their levels of motivation as a result of their relative half-time positions.

Finally, research on the “hot hand” theory also has relevance to the question of how success and failure may affect player motivation. A long history of research in basketball analytics beginning with Gilovich et al. (1985) and reinforced in later studies as summarized in Bar-Eli, et al. (2006) have maintained that “hot streaks” in successive shots is a fallacy. However, more recent research by Bocskocsky et al. (2014) has challenged these findings. Their spatial analysis of over 83,000 shots in the 2012-2013 National Basketball Association (NBA) season, shows that there is a small but significant increased likelihood of making a subsequent shot after success on a previous shot, once shot difficulty and defensive intensity have been controlled.

What is most important about the research of Bocskocsky et al. (2014) to the present study is not whether hot hand theory is a fallacy, but that both offensive and defensive players change their behavior after a made shot. Offensive players, they find, become more confident after making a shot and as a result are more likely to take the team’s next shot and to attempt more difficult shots (e.g., from farther from the basket). Their opponents, by contrast, are more likely to “turn up the heat” defensively after a made shot. Both of these findings are consistent with the hypotheses in the present research concerning overconfident winners and vengeful losers.

However the present research differs from these previous studies in several important ways. First it directly assesses the effects of previous game results on subsequent game outcomes for men’s college basketball on a large sample of regular season conference games. Second, it makes a significant methodological advance in that it controls the effects of “relative team strength” and other exogenous variables by not comparing one team to another but by comparing each team’s performance only to its own performance. In other words, we asked the question, what percentage of the time does a given team win (or lose) its subsequent game after winning (losing) its previous game? In so doing we are able to neutralize the effects of all exogenous variables. This methodological innovation is discussed more fully in the Method section below. Third, we examine the effects of previous game results not just on game outcomes but on a series of on-the-court performance variables such as shooting percentages, rebounds, and offensive efficiency. This analysis is essential for understanding the performance channels through which a previous win or loss can affect game outcomes.

This research approaches the question of previous game effects from a purely analytical perspective. It posits two intuitively logical and interrelated hypotheses regarding competitive motivations found in nearly all sports, and then tests them empirically using a highly robust game-level college basketball data base. The first is one we refer to as the “vengeful loser” hypothesis. It is based on the notion that a team that suffers a loss, especially a large loss or an embarrassing one, will channel the anger and emotion of that loss into a superior performance in its subsequent game. This hypothesis is about players collectively seeking to make up for a bad performance. It is driven by a desire to get back on track and achieve the goals they set for themselves, invariably goals that involve winning and championships. It simply states that the greater the loss in a previous game, the greater the likelihood of a victory in the subsequent game.

The second is a mirror image of the first; we call it the “over-confident winner” hypothesis. Players and fans know this when they see it. Teams become “cocky” and feel superior and infallible. Teams coming off a win, especially a large win, are thought to be “ripe for the picking” as they lack the emotional drive needed to prevail in a competitive matchup. The hypothesis states that the greater the win in a previous game, the greater the likelihood of a loss in the subsequent game.

While these two hypotheses may seem to ring true in a common sense, experiential way, we actually know very little about whether the effects are statistically real or illusory. Indeed, we note that these hypotheses run contrary to the possibility that past success could breed even greater success in sports. Vallerand and Losier (1999) posit that higher “intrinsic motivation” is promoted by success in sport and that failure will lead to lower intrinsic motivation. Moreover, if either of these opposing hypotheses is correct in its claim that game outcomes are influenced by previous results, in whichever direction, this then begs the question: are there measurable aspects of on-the-court performance, i.e., positive or negative play, which transmit these effects and result in the predicted outcomes? That is a question we come back to in the sections below.

Method

This analysis draws upon a data base of 5,048 NCAA Division I Men’s Basketball game pairs collected for three seasons: 2001-2002, 2010-2011, and 2011-2012. These are seasons for which the research team has complete season data available for analysis. There is nothing notable that sets these particular years apart from others in terms of the potential impacts of previous games on subsequent game results. Their spread over a 10-year period also gives us

greater confidence that any potential recent historical evolution of the game (e.g., growing importance of the 3-point shot) will also be captured in the data.

The data set for the sampled years contains all of the 111 teams in nine of the major college basketball conferences in the U.S.: Atlantic Coast (ACC), Atlantic 10, Big 12, Big East, Big Ten, Conference USA, Mountain West, Pac-12, and Southeastern (SEC). While the college basketball season normally runs from November through the end of the NCAA Tournament in early April, we focus exclusively on data from the conference regular season, which generally runs for a two-month period from late December to the end of February, before the conference tournaments start up in early March. During these months each team plays the other teams in its conference at least once but more often twice, depending on the size and structure of the conference. Teams typically play 16-18 games through this period. Conference scheduling also ensures that every team plays a balanced number of home and away games.

By targeting intra-conference games we avoid the biases and distortions observed during the non-conference season (November-December) where teams can be egregiously mismatched and invariably favor the major conference schools with home court/region advantage. Post-season games in March-April are likewise excluded from this analysis because the single-elimination structure of all of the conference tournaments and the NCAA Tournament means that only the winners advance to the next round, making these game-pairs unsuitable for the purpose of studying previous game effects. Other games dropped from the analysis are those very rare cases where the previous game for one of the teams was played more than one week earlier. With such a large gap between game pairs it is reasoned that the motivational effects of the previous game will have significantly waned.

Caution is required at this juncture to clarify that this research does not specifically employ techniques to measure individual player motivations in response to a previous win or loss, as might be done in an experimental design using a relatively small number of athletes and teams. Yet, what is lost in not experimentally measuring individual player motivations may be gained from the advantages of a large scale data base through which all other major known game influences can be randomized out by design across 5,048 game performances.

CONTROLLING FOR DIFFERENCES IN TEAM STRENGTH

A special challenge for the present analysis is to account for relative team strength; strong teams win more games than weaker teams and have higher winning percentages, regardless of whether they win or lose their previous games. Consequently, teams that win their previous games tend to win their subsequent games simply because they are better teams. To eliminate the team strength effect, we aggregate data in such a way as to compare each team's performance only to its own, proportionally across four categories of its previous game results:

1. Following a major loss (10+ points)
2. Following a minor loss (1-9 points)
3. Following a minor win (1-9 points)
4. Following a major win (10+ points)

A hypothetical example here may be instructive. Hoops University played 17 regular conference games; 16 of them, all but the first, followed a previous regular conference game. Of these 16 games, there were 4 in each of the 4 categories, (Loss 10+, Loss 1-9, Win 1-9, Win

10+). Hoops U won 100% of those in the first category, 75% in the second category, 50% in the third, and 25% in the fourth. We observe that Hoops U is more likely to win after a previous loss, especially after a big loss, than after a previous win. We note that the 10-point cut-off between a major win/loss and a minor win/loss is quite arbitrary, but settled on the 10-point cut-off for two reasons. The first is that in basketball there seems to be a convention that a "single digit" win or loss is a small one and a "double-digit" win is a large one. Second, we tested other cut-off levels (higher and lower) in our analysis and in all cases the resulting findings and conclusions were nearly identical to those reported here.

In computing these four winning percentages for all 111 teams, for each of the three study years, we generate an aggregated data set containing 1,189 cases at the "year-team-previous result" level. Analyses reported in the following section are conducted on this aggregate data base.

OTHERS COVARIATES

In addition to winning percentages for each of the four previous results categories, a set of 10 on-the-court performance variables is created for the same four categories. These variables constitute the eight principal performance variables recorded by the NCAA: field goal percentage, free throw percentage, total rebounds, assists, steals, blocks, turnovers and personal fouls, plus two computed variables, offensive and defensive efficiency ratings.

Offensive and defensive efficiency ratings are composite, team-level performance variables that are regularly used in basketball analytics as indicators of the overall quality of a team's play in an individual game or in the aggregate, across an entire season or more. They are also known as offensive points scored per possession (PPP), and defensive points per possession (i.e., the opponent's PPP). Team possessions (Oliver, 2004) are computed as:

$$\text{Possessions} = \text{FGA} - \text{OREB} + \text{TOV} + (0.4 * \text{FTA}), \quad (1)$$

where FGA=field goals attempted, OREB=offensive rebounds, TOV=turnovers, FTA=free throw attempts. Offensive and defensive efficiency ratings (PPP) are computed as:

$$\text{PPP} = \frac{\text{Points Scored}}{\text{Possessions}} * 100$$

Results and Discussion

WINNING PERCENTAGES BY PREVIOUS GAME RESULTS

Our analysis starts by testing the hypotheses that losing the previous game motivates teams to win their subsequent game, and, conversely, that winning the previous game instills overconfidence or a reduced sense of urgency and thus a tendency to lose the subsequent game. Table I reports the results of a one-way analysis of variance (ANOVA) comparing the mean win-

TABLE I
Impact of Previous Game Results on Game Winning Percentages and Performance

Game Results		Previous Game Results				Total	F	Sig.	R
		Lost by 10+ pts	Lost by 1-9 pts	Won by 1-9 pts	Won by 10+ pts				
Mean winning percentage		53.9	52.2	48.1	44.6	44.8	5.013	0.00**	-0.11**
Performance Variables	Mean field goal %	44.4	43.6	43.3	43.5	43.7	3.529	0.01*	-0.08**
	Mean free throw %	70.3	69.2	69.4	68.9	69.4	1.612	0.19	-0.05
	Mean total rebounds	31.3	31.4	31.4	31.2	31.3	0.265	0.85	-0.01
	Mean assists	13.5	13.4	13.3	13.1	13.3	0.881	0.45	-0.05
	Mean steals	6.5	6.4	6.4	6.2	6.4	1.087	0.35	-0.04
	Mean blocks	3.6	3.6	3.6	3.6	3.6	0.248	0.86	-0.01
	Mean turnovers	13.0	13.1	12.9	12.9	13.0	0.215	0.89	-0.02
	Mean personal fouls	18.3	18.1	18.1	18.2	18.2	0.535	0.66	-0.02
	Mean Off. Rating	104.4	102.2	101.9	102.1	102.6	3.854	0.00*	-0.07*
	Mean Def. Rating	102.3	102.5	102.2	104.5	102.9	4.083	0.00**	0.07*
(N)		(294)	(303)	(301)	(291)	(1189)			(1189)

** Significant at the 0.01 level (2-tailed).

* Significant at the 0.05 level (2-tailed).

ning percentages for teams in the four previous game results categories. As anticipated, teams that lose the previous game are significantly more likely to win the subsequent matchup, particularly when the previous loss is by 10 or more points (winning 53.9% of their games). If there were no previous game effect we would expect all four categories to win (and lose) exactly 50% of their subsequent games.

Similarly, the reverse hypothesis is supported by a like margin, with those having "won big" in the previous game coming up short in the subsequent game (winning only 44.6% of games). Comparing these means side by side, a team coming off a crushing loss is 22.6% more likely to win its next game than is the team coming off a blow-out win. From Figure 1 it can be seen that the relationship is fairly linear, though presumably there would be a leveling off at each tail if extended out to point differentials even greater than 10. In other words, a 30 point loss may not provide much more motivation to win the next game than does a 20 point loss. Figure 1 also confirms that the data support both study hypotheses in approximately equal fashion, with the positive effects of a previous loss proving comparable in magnitude to the negative effects of a win in the previous game.

We underscore that because we do not directly measure player motivation in this research the differences observed in relation to the results of previous games remain inferential. In other words, the data support the hypotheses put forward but they do not unequivocally prove them. It is the untested

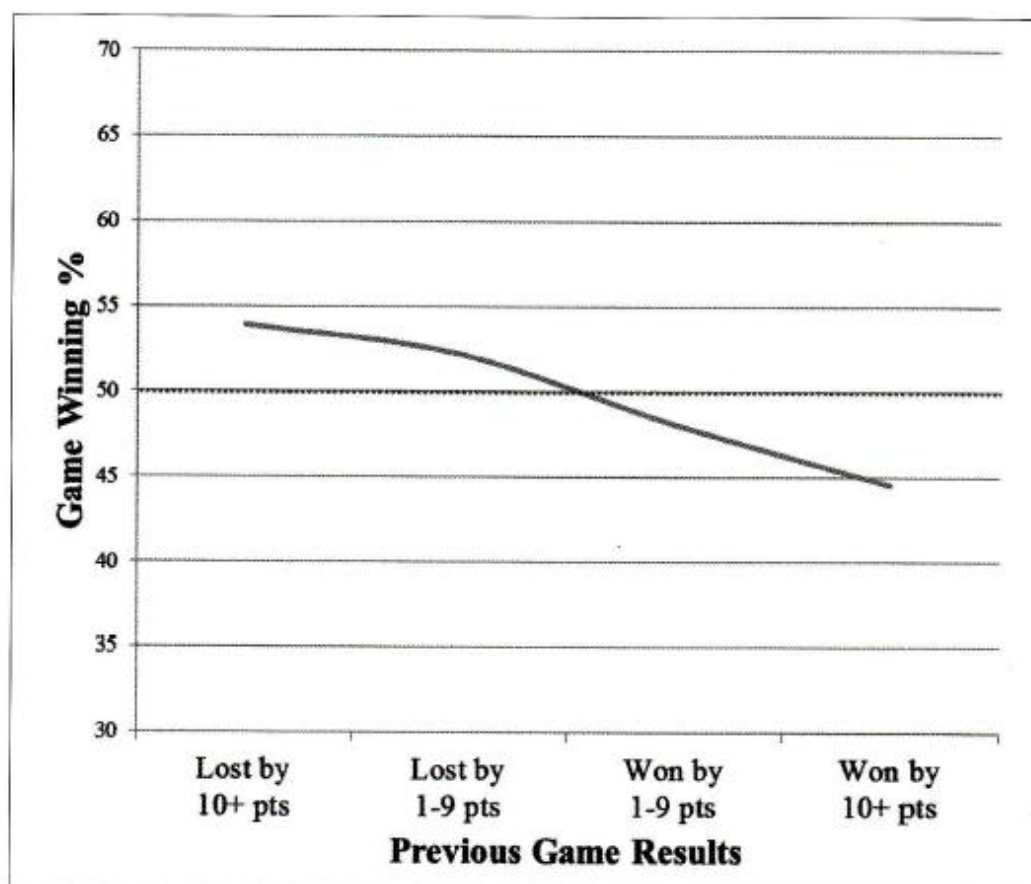


Fig. 1 - Mean Winning Percentage by Previous Game Results.

interpretation of the authors that the observed effects of previous games are most likely played out through how they differentially affect player motivation. Indeed, this must be seen as the most compelling account for why previous winners tend to lose and previous losers tend to win in subsequent games. It must also be recognized that motivations come from many sources pressure from fans, coaches, and of course internally, from the players themselves. How each of these motivational sources affects game outcomes, in light of a previous win or loss, remains a question for future research.

It is important to come back to a point made in the previous section, i.e., that the means presented in Table I reflect the average proportional success rates that teams achieve after wins and losses. And because only regular season conference games are included in the study population, we know that the number of home games is essentially equal to the number of away games and that the timing of home and away games as well as the strength of opposing teams played

are randomly distributed across the season. Thus both home court advantage and strength of team, by far the two most important determinants in basketball game outcomes, are randomized out by design. The proportional tally of wins and losses registered by each team in successive games throughout the season is consequently unbiased. And they have to be so, otherwise conference schedules would be seen as inherently unfair. That is not to say that in a particular year individual teams do not have easier or harder schedules than other teams, particularly in larger conferences where all teams do not play all other teams twice. But those advantages and disadvantages are random and are assumed to disappear when compared over multiple years, as we do in this study.

PERFORMANCE VARIABLES BY PREVIOUS GAME RESULTS

Knowing that game outcomes are influenced by previous results is an important first step. The second research question addressed in this study goes further; it asks whether there are particular channels through which these differences play out. That is, are there particular on-court performance variables that stand out as the more proximate, endogenous causes of success or failure? We test the effects of eight standard player performance variables: field goal percentage, free throw percentage, total rebounds, assists, steals, blocks, turnovers and personal fouls. Comparisons are also made for two composite, team-level performance variables, offensive and defensive efficiency ratings.

From the group of eight standard player performance variables only mean field goal percentage appears to be significantly affected by previous game results, where a previous loss is associated with a higher shooting percentage in the next game. But the results in Table I deserve a closer look. After field goal percentage, the remaining seven player performance variables all show a slight, though not statistically significant, improvement after a previous loss. This association can be observed both in the comparison of means and the correlation coefficients (*R*) reported. We surmise from the low yet remarkably consistent (100% predictable) pattern of correlations that the effects of a previous loss may in fact be channeled through the full array of on-the-court player performance variables. If there were no effect at all, one would expect truly random directionality in the set of correlations. Future research using a repeated measures design would help to substantiate this hypothesis.

Adding credence to this on-court performance interpretation are the Table I ANOVA results for mean offensive and defensive efficiency ratings. As described in the Method section, both of these variables are composite indicators comprised of several of the performance variables, and they consti-

tute an effective overall gauge of quality of play at the team-level. As visually demonstrated in Figure 2, and consistent with the pattern of findings across the individual player level performance variables, offensive efficiency ratings found to be are highest after a big loss, dropping to a steady lower level for all other previous game outcomes. Defensive efficiency shows a near mirror image pattern in which the rating is especially poor (high scores by the opponent) after a blow-out win, then improving (declining) for all other outcomes. Thus, we conclude that individually no performance variable stands out as a unique channel through which a previous loss or a previous win is expressed. Rather, these influences seem to affect in subtle ways both offensive and defensive performance ratings on the court and, in turn, final game outcomes.

Conclusion

Motivational factors are known to be among the leading determinants of game outcomes. Findings from this research strongly support the duel hypothesis that, in NCAA men's basketball, a previous loss tends to motivate a team to win its subsequent game, and that a previous win will in similar fashion increase the likelihood of a loss in the next game. We also conclude that the impact of a previous game is played out not through just one or two dimensions of on-the-court performance, but almost imperceptibly through numerous aspects of the game, such as shooting percentages, and offensive and defensive efficiency ratings. While the popular media often speaks of "momentum" being a factor, suggesting that teams get "on a roll" and build confidence with successive wins, our research shows compelling evidence to the contrary a win more often than not results in a subsequent loss.

This study provides us with some important lessons about the how previous successes and failures affect subsequent outcomes for college basketball, but it is possible that basketball is unique in this regard. It is widely known as a sport where emotions run high, where the flow of a game can hang on a single play, and where home court advantage is unparalleled in importance (Bois, 2011). How do previous game results play out in football, hockey and other team sports? And for how long do the effects last? Moreover, we focus in this study on only one subsequent game. But does the effect carry over to the next game? Does the previous game effect wane with a succession of losses to the point where teams give up and lose the emotion generated by a loss? It is in this longer-term perspective that questions about building team momentum, or falling into a "downward spiral" in a losing trend, may show greater validity and usefulness as research concepts.

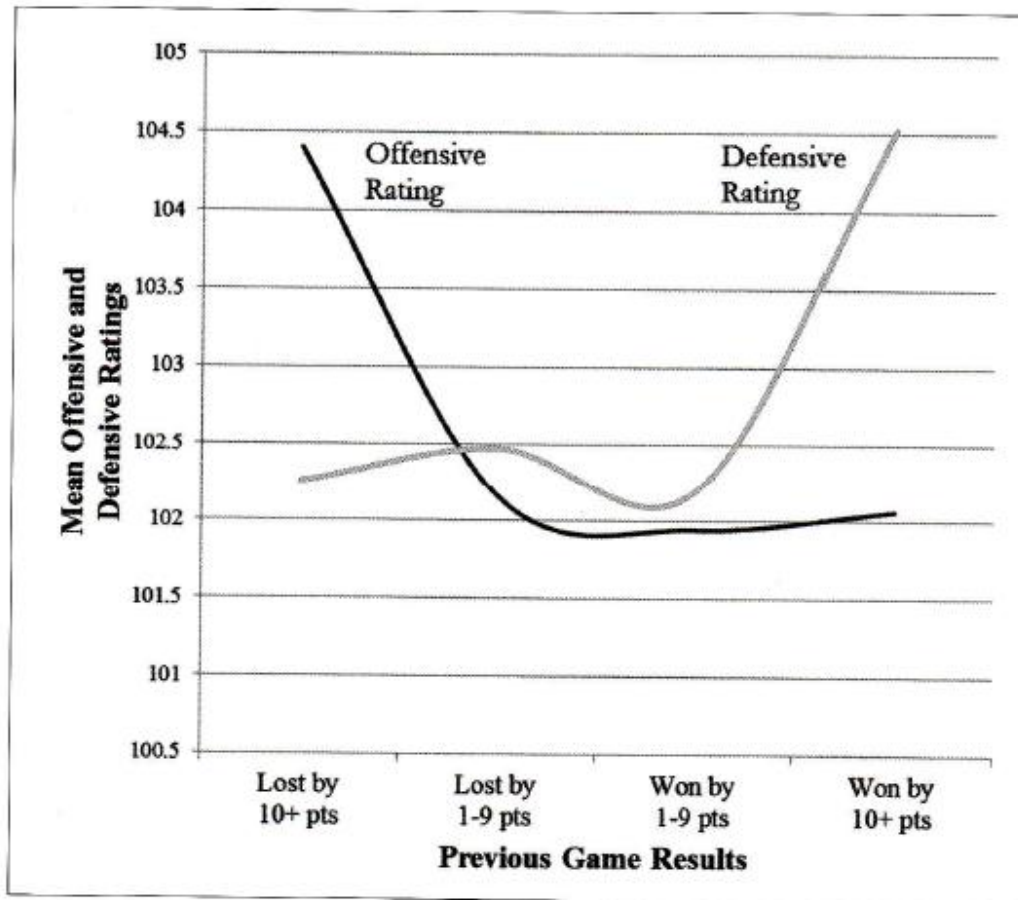


Fig. 2 - Mean Offensive and Defensive Ratings by Previous Game Results.

An additional, more practical implication stemming from the results of this research lies in what it all means for coaches and players as they prepare for a given game or opponent. The challenge of overcoming human nature as it is manifested in a subtle overconfidence from a "big win" is one that cannot be dismissed. The same goes for the challenge of preparing for a team that just suffered an embarrassing loss. While there can be no doubt that NCAA basketball coaches and players already have a basic awareness of these inherent dangers, and surely they recognize it when they see it in their own teams, it is evident that they are still underestimate what it takes to manage it. Putting the results of this study in their hands and in their heads will enable them to set previous game results alongside home court advantage as one of the critical ecological determinants of game outcomes; in doing so they will ultimately do a better job of managing both the vulnerabilities and opportunities that is creates.

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